

JONATHAN ONG CHONG AN

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Availability: Jan 2027 – Jul 2027

Singapore, SG

EDUCATION

Bachelor of Science, Information Systems (Product Development) Jan 2023 – Dec 2027
Singapore Management University | Second Major: Computer Science (Artificial Intelligence)

SUMMARY

AI and data engineering undergraduate with experience in enterprise GenAI, RAG, and production data systems development. Skilled in building reliable AI workflows with strong emphasis on grounding, testing, data quality, and responsible deployment.

EXPERIENCE

AI & Data Engineering Intern, Inland Revenue Authority of Singapore (IRAS) May 2026 – Nov 2026

- Migrated 10+ legacy SAS ETL scripts to PySpark for monthly batch-processing workflows handling approximately 5–10 million tax records, supporting downstream analytics and reporting.
- Evaluated an internal GenAI assistant across 100+ UAT and adversarial test scenarios, identifying unsafe, ungrounded, and irrelevant responses and supporting improvements that achieved the required 90% safety-score threshold.
- Enhanced internal AI assistant's retrieval workflow through integration with internal search APIs, improving access to trusted organisational knowledge and supporting more relevant, evidence-grounded responses.
- Supported the migration of internal machine-learning APIs to Azure ML by standardising model artefact management, strengthening logging and exception handling, and introducing automated pytest checks that increased the CI test success rate by approximately 50%.

DevOps Engineer Intern, YTL PowerSeraya Jan 2026 – Apr 2026

- Built and rolled out Azure DevOps multi-stage YAML pipelines for 4 internal gas management Azure applications, reducing deployment time from 60 to 20 minutes and improving release consistency.
- Designed a self-hosted agent approach for secure IaaS deployments within an isolated VNet, enabling CI/CD execution on application VMs and documenting the workflow for developer adoption.

Cloud Engineer Intern, Global Enterprise Mobility (GEM) May 2025 – Aug 2025

- Built Jenkins CI/CD pipelines using Terraform-provisioned AWS infrastructure and GCP Secret Manager, reducing release time from 40 to 20 minutes while improving deployment repeatability and credential security.
- Deduplicate webhooks with a DynamoDB idempotency layer, resolving production defect affecting 12–15% of transactions.

PROJECTS

PitchPerfect — AI Evaluation Platform | Product Owner & AI Data Pipeline Lead Jun 2026 – Present

- Product Owner in a six-member Agile Scrum team, collaborating with the Singapore Management University Institute of Innovation and Entrepreneurship to develop AI Evaluation tool for the annual Lee Kuan Yew Global Business Plan Competition.
- Own the ingestion and retrieval workstream for an AI-assisted pitch-deck evaluation system, translating an evaluator's manual review process into structured feature extraction, evidence retrieval, and readiness assessment stages.
- Designing a pipeline to ingest labelled pitch decks, extract rubric-aligned features and metadata, generate embeddings, and retrieve supporting evidence for consistent evaluation.

YilongMa — RAG Data Pipeline | SMU Natural Language Processing Project Aug 2025 – Dec 2025

- Built an end-to-end RAG pipeline using ChromaDB, Llama 3.2 1B, and hybrid retrieval to generate grounded responses from a transcript-based knowledge base.
- Processed more than 50 hours of audio using Whisper and developed evaluation notebooks to assess retrieval relevance and grounded-response quality.

Wander — AI-Powered Travel Itinerary | SMU Full-Stack Project Sep 2024 – Nov 2024

- Built a full-stack travel planning application using Vue.js and Flask, integrating the OpenAI API to generate personalised itineraries based on user preferences.
- Implemented dynamic location suggestions from submitted TikTok URLs, contributing to an A+ project grade.

CERTIFICATIONS

AWS Certified Cloud Practitioner | Google IT Automation with Python | Java Certified Foundations Associate

SKILLS

Languages Frameworks: Python, SQL, Bash, JavaScript, Java, Vue.js, React.js

AI/ML: RAG, embeddings, hybrid retrieval, Prompt Engineering, LLM evaluation, OpenAI API

Data Engineering: PySpark, Pandas, Azure Synapse, Databricks, SAS migration

Software/Cloud: FastAPI, Pytest, Docker, Terraform, Jenkins, Azure Pipelines, Git, AWS, Azure, Google Cloud Platform

Databases: PostgreSQL, DynamoDB, Amazon RDS, Azure Data Lake, PgVector, ChromaDB